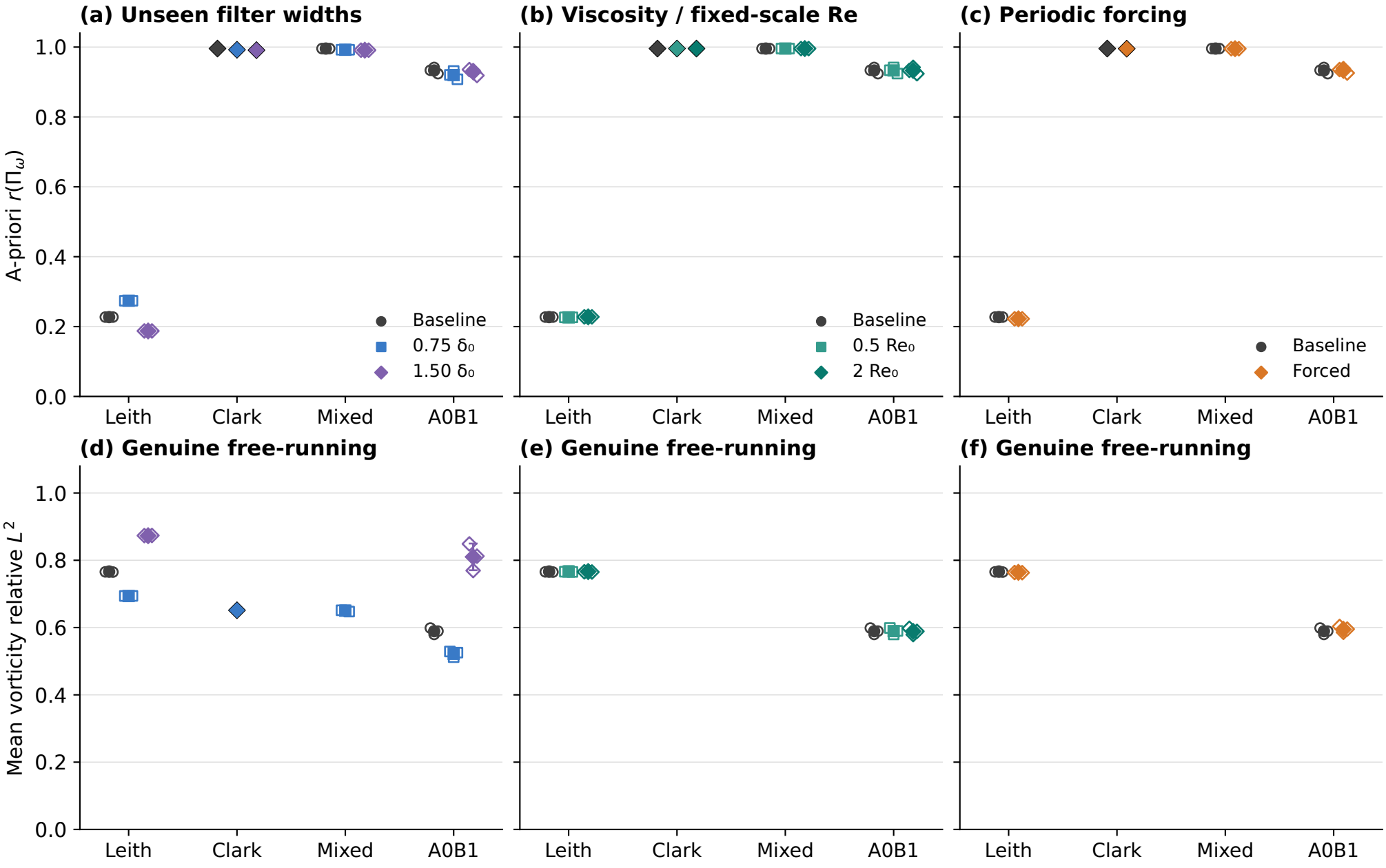
Classical versus neural SGS behavior under unseen physical regimes
(a) Unseen filter widths
(b) Viscosity / fixed-scale Re
(c) Periodic forcing
(d) Genuine free-running
(e) Genuine free-running
(f) Genuine free-running
A-priori r(Π_ω)
Mean vorticity relative L^2
1.0
0.8
0.6
0.4
0.2
0.0
Leith
Clark
Mixed
A0B1
Baseline
0.75 δ₀
1.50 δ₀
0.5 Re₀
2 Re₀
Forced